

The indications for BCT or mastectomy and its sequencing with chemotherapy are described in the previous chapter on management algorithms. Indications for special surgical procedures are described here.

Sentinel node biopsy and Axillary sampling: Several RCTs^{28,29} show that in women with small tumors and clinically negative axilla, the morbidity from axillary dissection done as a therapeutic or staging modality can be avoided by careful histological examination of the sentinel lymph nodes identified by various methods and conducting full axillary dissection only in those with positive sentinel nodes. Sentinel Node Biopsy (SNB) is performed using 0.3ml sub-dermal injection of patent blue dye or 2% methylene blue 10 minutes prior to the surgical incision and 0.5ml of technetium-labeled sulphur colloid 2 hours prior to surgery. The technique should be validated in each centre for its sensitivity and specificity before offering it as a routine. Low axillary sampling, an alternative to SNB can be practiced more widely in developing countries. If histological evaluation of sentinel lymph nodes or low axillary sampling does not reveal any metastasis, completion axillary clearance is not warranted. However, if the SNB is positive or the paraffin sections show metastasis, completion axillary clearances (Level I – III) are generally considered. There is however recent EORTC-AMAROS RCT showing that in SNB + ve cases, axillary RT achieves axillary control rates similar to completion axillary clearance but with different morbidity pattern (Donker et al, Lancet Oncol. 2014).

Breast reconstruction: In centres with such expertise, breast reconstruction should be offered to women who are likely to have a poor cosmetic outcome after BCT or are undergoing mastectomy and not at excessive risk of local recurrence or wound complication. Because of the resource constraints and optional nature of this service, few centres in the country, especially in the busy public hospitals, routinely offer or perform breast reconstructions for majority of the patients who may benefit from such reconstruction. Hence good quality breast reconstruction services may be prioritized.

Whole breast reconstruction (WBR) with or without nipple areolar reconstruction should be offered to motivated young women who have to undergo a mastectomy for a DCIS or early breast cancer (for reasons such as diffuse micro-calcification or multi-centric disease) or as a prophylactic surgery as in BRCA1 mutation carriers. Reconstruction could be immediate or delayed. WBR can be done with **pedicle flap** (Latissimus dorsi flap with implant or expander or a Transverse Rectus Abdominis Myocutaneous flap) or as a **Free flap with microvascular anastomosis** (Deep Inferior Epigastric perforator flap; Anterolateral thigh flap or Superior Gluteal artery perforator flap)

Partial breast reconstruction (PBR): Is indicated after upfront surgery for large tumor with micro-calcification or after NACT with unfavourable breast to tumor ratio and patient is keen on breast conservation. Reconstruction can be done using autogenous tissue (fat, muscle, skin and vasculature) from the back (latissimus dorsi flap), abdomen or thigh as a free or pedicled flap.

Optimal management of breast cancers hinges upon standardized histopathology evaluation and reporting³⁰.

All specimens should be transported to the pathology lab immediately. In case a delay is expected surgeons should slice specimen's posterior aspect upwards.

All lumpectomies should be treated as potentially harboring cancer and should be oriented for margin evaluation. A six surface margin (anterior, posterior, inferior, superior, medial and lateral) is sufficient for routine evaluation though more extensive sampling may result in greater margin positivity. Close margins are to be evaluated radially whereas shave margins should be evaluated for further margins.

All specimens should be fixed in 10% neutral buffered formalin with pH 7.2 to 7.4. Sections for ER/PR/Her2neu estimation should be fixed for no less than 6 hours and no more than 72 hours before processing. ASCO-CAP guidelines suggest recording the time specimen is out of patients body and time specimen is fixed in formalin for validating specimen fixation. ASCO-CAP guideline state "It is the responsibility of the surgeon and operating room staff or the radiologist and his/her staff obtaining the specimen to document the collection time, and it is the responsibility of the pathologist and laboratory staff to document the fixation start time".

For lymph nodes, record the number, size of largest node and number of grossly metastatic nodes. One section from grossly metastatic nodes is enough. For lymph nodes >5mm in maximum size slice at approximately 3mm or less perpendicular to the long axis. Lymph nodes <5mm should ideally be bisected and blocked; alternatively, lymph nodes <5mm can be blocked in their entirety. Nodes should be counted even on microscopy to avoid duplication of nodes during submission.

For sentinel node, slice at 2mm interval and submit entirely. If the initial H&E stained section is negative, four levels are cut. Three of these sections are stained by H&E and one randomly chosen section is stained with an epithelial IHC stain.

Sections to be taken

Lumpectomy

A: Four sections from tumor all with adjacent normal breast;

B to G: Six surface margins;

H: Remaining Axillary /apical nodes

Mastectomy

Specimens are evaluated similarly but base is the only relevant margin.

Nipple areola and overlying skin should be evaluated whenever included in specimen.

All mammographically detected lesions should be grossed under mammographic guidance. Make sure to block the entire lesion so as that the size of lesions can be reconstructed microscopically. Also sample margins of lesions radiologically.

Essentials in a Histopathology report:

- Tumor size (all 3 dimensions)
- Tumor type
- Tumor grade (Modified Richardson Bloom Score)
- Presence of extensive intraductal carcinoma (EIC)*
- Lymphovascular emboli
- Margin status (negative /close /focal positive /gross positive) in wide excision**
- No. of metastatic nodes and total axillary lymph node dissected
- Receptor status: ER and PR (by IHC or EIA) and HER2neu by IHC. ER and PR should be scored as described later.**
- HER2neu of 2+ on IHC is considered equivocal and this can be resolved by FISH test for gene amplification, which is considered amplified or positive if score >2.2.

***EIC:** DCIS in >25% of any low power field within or outside the tumor. It is a strong predictor of local recurrence after BCT, especially if margins are positive.

****Gross +ve cut margin:** Extensive involvement of a cut margin OR >3 foci of invasive or in-situ carcinoma in any inked margin. It requires revision excision or mastectomy.

****Focal positive cut margin:** 3 or less foci of invasive or in-situ carcinoma in any inked margin. It requires revision surgery especially if EIC is present

Modified Richardson Bloom grading system (Nottingham's modification)

	Tubule	Nuclear grade	Mitosis/10hpf*
Score1	>75%	Small uniform nuclei, no pleomorphism	0-6
Score2	10-75%	Moderate sized nuclei with pleomorphism	7-12
Score3	<10%	Marked pleomorphism	>12

Final score 3-5=Grade I, Final Score 6-7=Grade II, Final Score 8-9=Grade III

*Could vary with microscope.

- Grade all carcinomas, including post NACT, special types and core biopsies.
- ER/PR estimation should be routinely performed. ER/PR expression in >1% of cells is reported as positive for hormone receptors. For ER/PR estimation it is crucial to fix tissues in 10% neutral buffered formalin and employ pressure cooking methods for antigen retrieval. When reporting on outside referral material, if fixation procedure is not documented, poor quality of fixation, if evident, should be commented. Though ASCO-CAP guidelines suggest that such samples should be rejected, it is not feasible in many Indian practice settings. Including normal breast tissue along with tumor section helps to assess internal control for hormone receptor and validate a given test result^{31,32}.

- Her2neu IHC testing should ideally be done by the FDA approved Hercep test OR Ventana Her2neu assay. However these FDA approved assays are not commonly used in India due to their cost. Hence before using any non validated antibody in routine practice, the laboratory should validate it either with FISH or the FDA approved IHC assays. Any Her2neu IHC showing membranous staining of normal breast epithelium should be reevaluated. Since patients with IHC score 3 may receive Trastuzumab without FISH confirmation, strict guidelines for interpretation and reporting should be followed. Only a complete and crisp membrane staining without any cytoplasmic staining in >30% of cells should be interpreted as score 3 or positive. Score 2 staining results in complete membrane staining but in <30% of cells and considered equivocal. All other staining is negative (score 1/0). For FISH, count non overlapping cells and select block for invasive tumor. Reject FISH if signals are non-uniform, background obscures interpretation, autofluorescence³³.
- Ki67 correlates with tumour grade and can provide useful additional prognostic information in low grade tumours. Validated Ki67/MIB1 testing in node negative, small, low grade tumours can help decide the need for chemotherapy in such cases.